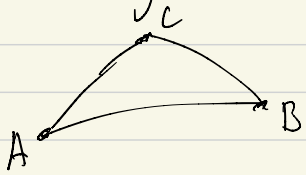
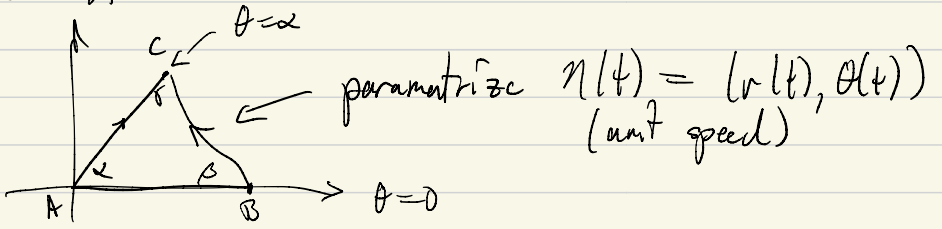
Geodesic triangles

Consider a triangle T with all sides geodesic



We will assume T fits in a geodesic polar patch centred at



With $ds^2 = dr^2 + \phi^2 d\theta^2$, the geodesic equations become

$$\frac{d}{dt}(\dot r) = \phi(\partial_r \phi)\cdot \dot\theta^2$$

$$\frac{d}{dt}(r^2\dot\theta) = \phi(\partial_\theta\phi)\cdot \dot\theta^2$$

define an orthonormal frame $e_1 = \partial_r$, $e_2 = \frac{1}{\phi}\partial_\theta$

Let ψ = angle from e_1 to $\dot\eta$ (>0 since η is not radial)



Since $\dot\eta$ is a unit vector,

$$\dot\eta = \cos\psi\, e_1 + \sin\psi\, e_2$$

Compare this to chain rule $\dot\eta = \dot r\,\partial_r + \dot\theta\,\partial_\theta$

$$\longrightarrow \dot r = \cos\psi\,,\quad \dot\phi = \frac{\sin\psi}{\phi}$$

Then

$$\frac{d}{dt}(\cos\psi) = \dot r'$$

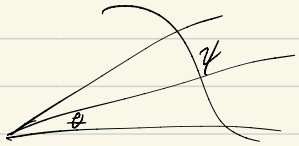
$$-\sin\psi\cdot\dot\psi = \phi\,\dot\theta^2\,\partial_r\phi \qquad \text{(geodesic eq.)}$$

$$= \sin\psi\cdot\dot\theta\,\partial_r\phi$$

$$\dot\psi = -\dot\theta\,\partial_r\phi$$

Conclusion: along $\mathcal{H}$,

$$\frac{d\psi}{d\theta} = -\partial_r \phi$$



Gauss-Bonnet Thm (triangle version)

For a geodesic triangle within a geodesic coord. patch,

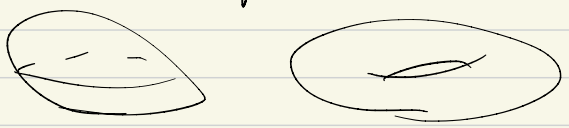
$$\iint_T K \, dA = (\alpha + \beta + \gamma) - \pi$$

Proof:

$$\iint_T K \, dA = \int_0^{\alpha} \int_0^{r(\theta)} \underbrace{\left(-\frac{\partial_r^2 \phi}{\phi} \right)}_{K} \cdot \underbrace{\phi \, dr \, d\theta}_{dA}$$

Global version

Let S be a compact surface without boundary



etc.

Claim: by covering S with geodesic patches, we can subdivide S into geodesic triangles



$S = \cup T_j$ $\qquad \int_{T_j} K\,dA = (\alpha_j + \beta_j + \theta_j) - \pi$

For the triangulation $\{T_j\}$ let $v = \#$ vertices
$e = \#$ edges
$f = \#$ faces

(no boundary $\Rightarrow 3f = 2e$)

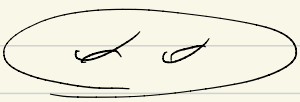
Summing over faces gives

$$\int_S K\,dA = \sum_{j=1}^{f} (\alpha_j + \beta_j + \theta_j - \pi)$$
$$= 2\pi v - \pi f$$
$$= 2\pi(v - e + f)$$

Euler characteristic $\chi(S)$ (independent of triangulation)

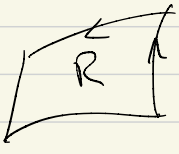
Gauss-Bonnet Thm (cpt surface)

For a compact surface S, $\int K\,dA = 2\pi\chi$



genus $g = \#$ of holes
$\chi = 2 - 2g$

Full version: on an oriented surface S, let R be a cpt region with piecewise smooth boundary ∂R



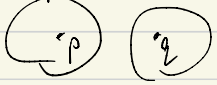
$$\int_R K\,dA + \int_{\partial R} k_g\,ds = 2\pi\chi - \Sigma(\text{exterior angles})$$

Smooth Manifolds

A smooth manifold M is a 2nd countable Hausdorff space equipped with a smooth atlas.

2nd countable: The topology of M is generated by a countable collection of open sets

Hausdorff: Any 2 pts. are contained in disjoint open sets



smooth atlas: M is covered by open sets U_j with homeomorphisms $\sigma_j : V_j \subset \mathbb{R}^n \to U_j$ so that transition maps are smooth on overlaps

(Whitney Embedding Thm: without loss of generality, we can assume $M \subset \mathbb{R}^{2n}$)